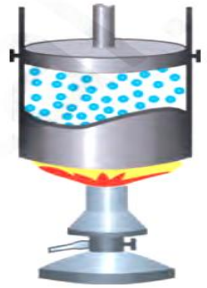
Fizika

9-sinf



Mavzu:

Termodinamikaning birinchi qonuni.

Energiyaning saqlanish qonuni

Tabiatda energiya yo‘qdan bor bo‘lmaydi va yo‘qolmaydi. Energiya miqdori o‘zgarmaydi, energiya faqat bir turdan boshqa turga aylanadi.

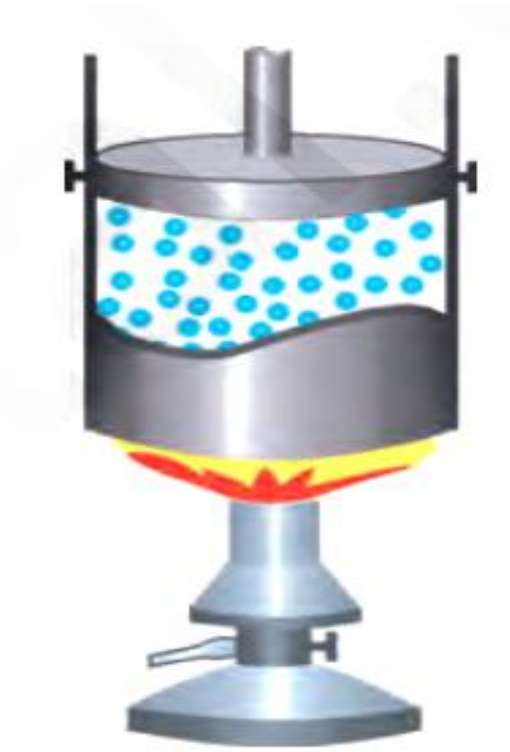
Energiyaning saqlanish qonuni tabiatda sodir bo‘ladigan barcha hodisa va jarayonlarda bajariladi. ***Termodinamikaning birinchi qonuni*** energiyaning saqlanish qonunini issiqlik jarayonlariga tadbiqini ifodalaydi.

Termodinamikaning birinchi qonuni

$$Q = \Delta U + A$$

Sistemaga berilgan issiqlik miqdori sistemaning ichki energiyasini o‘zgartirishga va tashqi kuchlarga qarshi ish bajarishga sarflanadi.

Bu qonunni XIX asr o‘rtalarida R. Mayer, G. Gelmgols va J. Joul ta’riflagan.

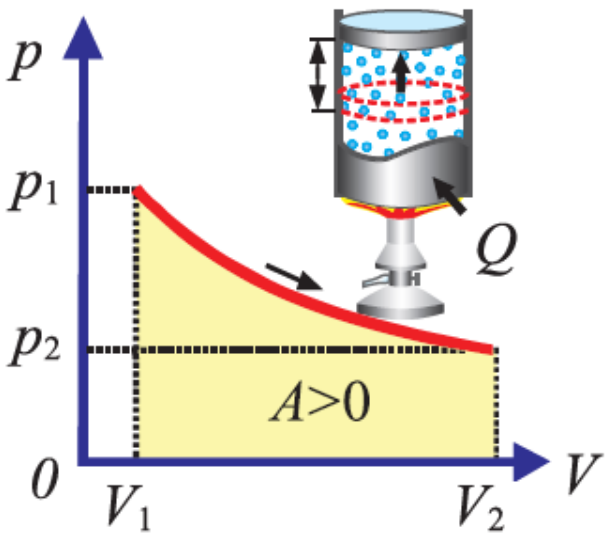


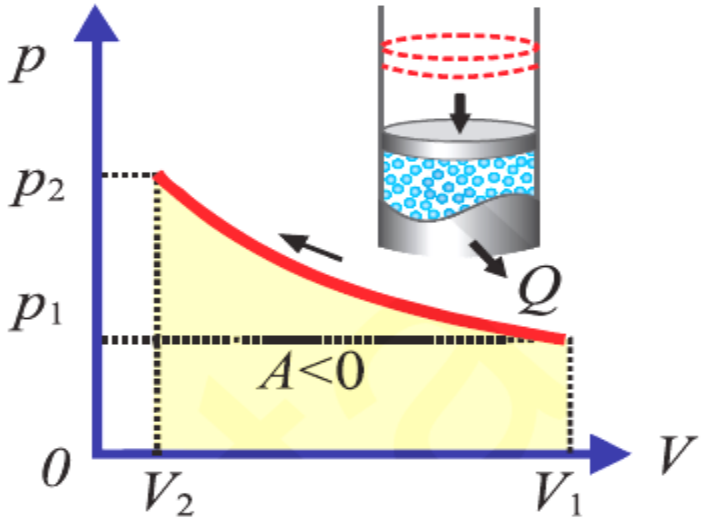
Termodinamikaning birinchi qonunini izojarayonlarga tadbiqi: *Izotermik jarayon*

$T = const.$ $Q = \Delta U + A$ dan $\Delta T = 0$ da $\Delta U = 0$

$$Q = A \qquad A = p\Delta V = \nu R \Delta T$$

Izotermik jarayonda gazga berilgan issiqlik miqdori ish bajarishga sarflanadi.

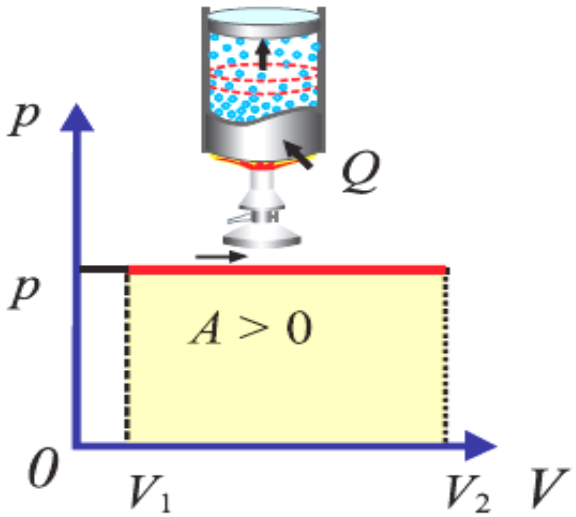


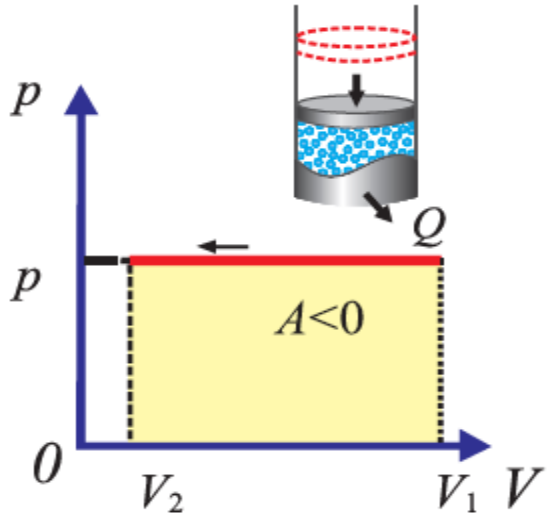


Izobarik jarayon (p= $const$.)

$$Q = \Delta U + A \qquad \Delta U = \frac{3}{2}\nu R\Delta T \qquad A = p\Delta V = \nu R\Delta T$$

Izobarik jarayonda sistemaga berilgan issiqlik sistemaning ichki energiyasini o‘zgartirishga va o‘zgarmas bosimda ish bajarishga sarflanadi.





Izoxorik jarayon $(V = const.)$

$Q = \Delta U + A$ dan $A = p\Delta V$ $\Delta V = 0$ dan $A = 0$

$$Q = \Delta U \qquad \Delta U = \frac{3}{2}\nu R\Delta T$$

Izoxorik jarayonda sistemaga berilgan issiqlikning hammasi ichki energiyani o‘zgartirishga sarflanadi.

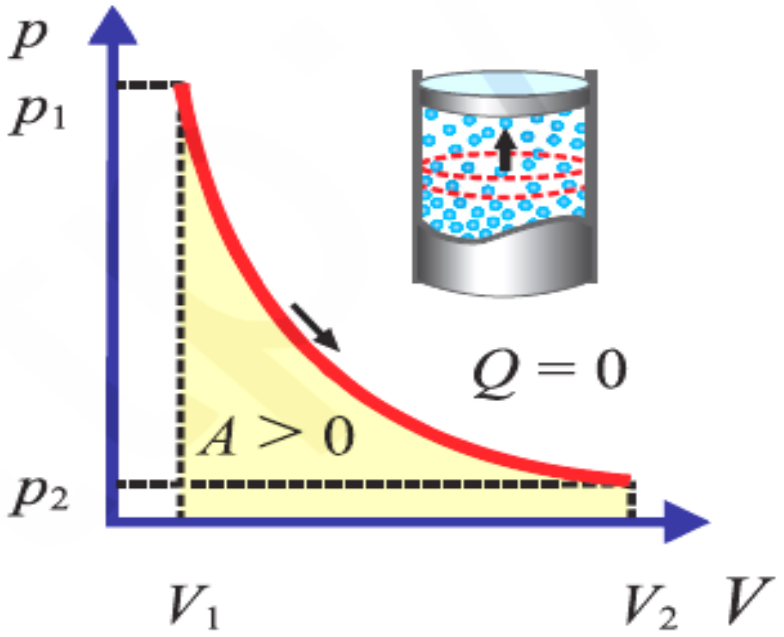
Gaz isitilganda ichki energiyasi ortadi, sovitilganda esa kamayadi.

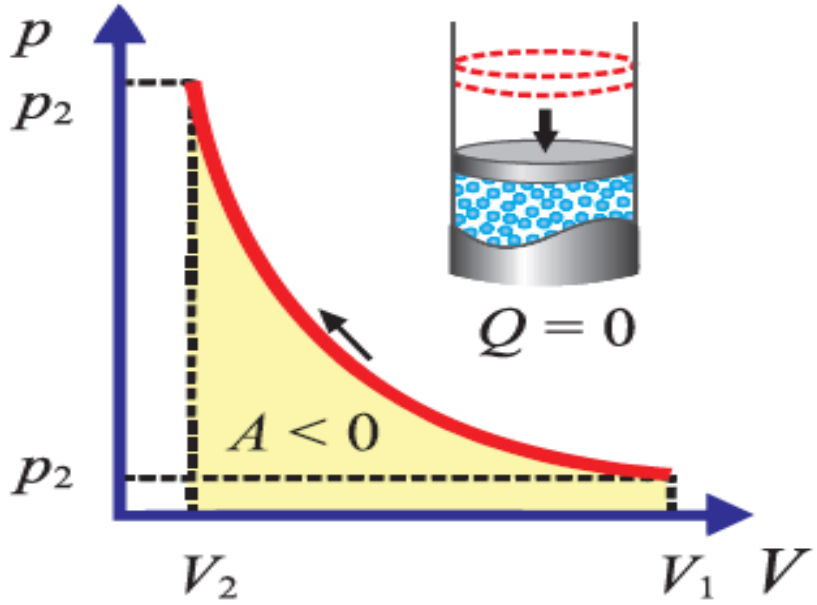
Adiabatik jarayon $(Q = 0)$

Issiqlik almashinmaydigan qilib izolyatsiyalangan sistemadagi jarayonga ***adiabatik jarayon*** deyiladi.

$$Q = \Delta U + A = 0 \qquad \Delta U = -A$$

Adiabatik jarayonda gazning 3 ta makroskopik parametrlari bosimi (p), hajmi (V) va temperaturasi (T) o‘zgaradi.





Masala

Gaz qizdirilganda uning ichki energiyasi 400 J dan 700 J gacha ortdi. Agar gazni qizdirish uchun 1000 J issiqlik sarflangan bo‘lsa, bunda bajarilgan ishni toping.

Berilgan:

$U_1 = 400\ \text{J}$

$U_2 = 700\ \text{J}$

$Q = 1000\ \text{J}$

Topish kerak: $A - ?$

Formula:

$$Q = \Delta U + A$$

$$A = Q - \Delta U$$

$$\Delta U = U_2 - U_1$$

$$A = Q - (U_2 - U_1)$$

Yechish: $A = 1000\ J - (700\ J - 400\ J) = 700\ J$

Javob: $A = 700\ \text{J}$

Masala

Izobarik qizdirishda gazga 16 J issiqlik berildi, natijada gazning ichki energiyasi 8 J ga, hajmi esa $0{,}002\,m^3$ ga ortdi. Gazning bosimini (kPa) da toping.

Berilgan:

$Q = 16\ \text{J}$

$\Delta U = 8\ \text{J}$

$\Delta V = 0{,}002\ m^3$

Topish kerak: $p - ?$

Formula:

$$A = p\Delta V \qquad p = \frac{A}{\Delta V}$$

$$Q = \Delta U + A$$

$$p\Delta V = Q - \Delta U \qquad p = \frac{Q - \Delta U}{\Delta V}$$

Yechish: $p = \frac{16\ J - 8\ J}{0{,}002\ m^3} = 4000\ Pa = 4\ kPa$

Javob: $p = 4\ kPa$

Mustaqil bajarish uchun topshiriqlar

1. Mavzuni o‘qish, mavzuga doir savollarga javob yozish. (70-bet)

2. Gaz izoxorik qizdirilganda uning ichki energiyasi 200 J dan 300 J gacha ortdi. Gazga qancha issiqlik berilgan?

3. Uyingizdagi termosni kuzatib, u qanday qismlardan tashkil topganini o‘rganing. Nima uchun termosda uzoq vaqt harorat o‘zgarmay turishi haqidagi fikringizni daftaringizga yozing.